



PA31 Quick Reference Handbook (QRH)

Emergency/Abnormal Procedures and System Checks

EVACUATION DRILL		(1A)
<u>'CAPTAINS DUTIES'</u>	<u>'F/Os DUTIES'</u>	
CALL "EVACUATE LEFT OR RIGHT"		
BOTH FIREWALL FUEL SHUTOFFS - OFF	ADVISE ATC INCLUDING SOULS ON BOARD, FUEL LOAD AND D.G.s.	
BOTH FUEL SELECTORS - OFF		
BOTH EMERGENCY FUEL PUMPS – OFF		
BOTH FUEL BOOST CB'S - PULLED	TURN ON ENTRANCE LIGHT IF REQUIRED	
BOTH MIXTURES - ICO		
BOTH THROTTLES - CLOSED		
BOTH MAGS - OFF	USE PORTABLE FIRE EXTINGUISHER IF REQUIRED AND COMMENCE EVACUATION	
MAKE RADIO CALL (IF SINGLE PILOT)		
MASTER – OFF		
ASSIST WITH EVACUATION AND CONFIRM THAT ALL PERSONS ARE ACCOUNTED FOR	DIRECT AND ASSIST PASSENGERS AS APPLICABLE	

ENGINE SECURING PROCEDURE		(1B)
1)	POWER ON OPERATING ENGINE	SET
INOPERATIVE ENGINE ONLY		
2)	THROTTLE	CLOSE
3)	PROPELLER (1000 RPM MIN)	FEATHER
4)	MIXTURE	ICO
5)	COWL FLAPS	CLOSE
6)	FIREWALL FUEL SHUTOFF	OFF
7)	FUEL SELECTOR	OFF
8)	EMERGENCY FUEL PUMP	OFF
9)	MAGS	OFF
10)	AIRCONDITIONER	OFF
11)	FUEL BOOST CB	PULLED
12)	ALTERNATOR CB	OFF
13)	ELECTRICAL LOAD	REDUCED
14)	CROSSFEED	CONSIDER (5A)

REJECTED TAKEOFF WITH SUFFICIENT RUNWAY TO STOP		(1C)
1)	THROTTLE	IDLE
2)	BRAKES	AS REQUIRED
AFTER AIRCRAFT HAS COME TO A STOP, SET PARKING BRAKE, AND FOLLOW THE APPLICABLE DRILL OR ABNORMAL PROCEDURE AS APPLICABLE		
NOTE IF THERE IS A FIRE THE EVACUATION DRILL WILL BE CALLED.		



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REJECTED TAKEOFF WITHOUT SUFFICIENT RUNWAY TO STOP		(2A)
1)	THROTTLES	IDLE
2)	BRAKES	MAX – Steer to avoid obstacles
3)	MIXTURES	ICO
4)	FIREWALL FUEL SHUTOFFS	OFF
5)	EMERGENCY FUEL PUMPS	OFF
6)	MAGS	OFF
7)	MASTER	OFF
AT FULL STOP		
8)	EVACUATION DRILL	COMPLETE (1A)

ENGINE FAILURE AFTER ROTATION (TAKEOFF CONTINUED)		(2B)
1)	POWER	MAXIMUM
2)	GEAR	UP
3)	FLAPS	UP
4)	PROP (INOP ENGINE)	FEATHER
5)	AIRSPEED	MAINTAIN AS ABLE TARGET 104 KTS
6)	DIRECTIONAL CONTROL	MAINTAIN 5° INTO OPERATING ENGINE
IF THERE IS AN ENGINE FIRE		
7)	FIREWALL FUEL SHUTOFF (INOP ENGINE)	OFF
8)	COWL FLAPS (INOP ENGINE)	CLOSE
9)	AIRSPEED (CLEAR OF OBSTACLES)	107 KTS
10)	TRIM (5 DEGREES TO GOOD ENGINE)	ADJUST
11)	COWL FLAPS (OPERATIVE ENGINE)	AS REQUIRED
12)	ENGINE SECURING PROCEDURE	COMPLETE (1B)

ENGINE FIRE AFTER ROTATION (OR AFTER DECISION SPEED WITH DECISION TO CONTINUE)		(2C)
1)	POWER	MAXIMUM
2)	GEAR	UP
3)	FLAPS	UP
4)	OBSTACLES	CLEAR
5)	AIRSPEED	107 KTS
AT 400 FEET		
6)	FIREWALL FUEL SHUTOFF (INOP ENGINE)	OFF
7)	PROP (INOP ENGINE)	FEATHER
8)	COWL FLAPS (INOP ENGINE)	CLOSE
9)	ENGINE SECURING PROCEDURE	COMPLETE (1B)



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ENGINE FAILURE/FIRE IN FLIGHT (AT OR ABOVE SAFE ALTITUDE)		(3A)
1)	INOP ENGINE	IDENTIFY
2)	OPERATIVE ENGINE	ADJUST POWER TO MAINTAIN ALT AND AIRSPEED
3)	AIRSPEED MAINTAIN	107 KTS PLUS
4)	GEAR	UP
5)	FLAPS	UP
IF THERE IS A FIRE SKIP CAUSE CHECK. PROCEED TO STEP 13		
IF NO FIRE COMPLETE STEPS 6 TO 12 BEFORE SECURING INOP. ENGINE		
6)	FUEL PRESSURE AND FLOW	CHECK. IF PRESSURE <34 PSI, SELECT EMERGENCY FUEL PUMP ON
7)	FUEL QUANTITY	CHECK
8)	FUEL SELECTOR	SWITCH TO OPPOSITE TANK IF FUEL REMAINING
9)	OIL PRESS/ TEMP	CHECK
10)	MAGS (LEFT AND RIGHT FOR FAILED ENGINE)	ON INDIVIDUALLY
11)	MIXTURE	ADJUST
12)	AIR START	CONSIDER (6B)
IF ENGINE DOES NOT START, COMPLETE ENGINE SECURING PROCEDURE (1B)		
13)	FIREWALL FUEL SHUTOFF (INOP ENGINE)	OFF
14)	PROP (INOP ENGINE)	FEATHER
15)	FUEL PUMP (INOP ENGINE)	OFF
16)	COWL FLAP (INOP ENGINE)	CLOSED
17)	POWER (OPERATIVE ENGINE)	POWER AND MIXTURE AS REQUIRED
18)	FUEL (OPERATIVE ENGINE)	FUEL QUANTITY AND EMERGENCY FUEL PUMP AS REQUIRED
19)	COWL FLAP (OPERATIVE ENGINE)	AS REQUIRED
20)	TRIM (5 DEGREES TO GOOD ENGINE)	ADJUST
21)	ELECTRICAL LOAD	DECREASE TO MINIMUM
22)	ENGINE SECURING PROCEDURE	COMPLETE (1B)

ENGINE FIRE DURING START		(3B)
1)	CONTINUE CRANKING	
2)	BOOST PUMPS	OFF
3)	MIXTURE	IDLE CUT OFF
4)	THROTLES	OPEN
IF FIRE PERSISTS		
5)	FIREWALL FUEL SHUTOFF	OFF
6)	MASTER	OFF
7)	EVACUATION DRILL	COMPLETE (1A)



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ENGINE FIRE DURING GROUND OPERATIONS		(4A)
1)	BRAKES	APPLY AND SET
2)	EVACUATION DRILL	COMPLETE (1A)
NOTE APPLIES TO ALL ENGINE FIRES FROM ENGINE START TO REJECTED TAKEOFF WITH SUFFICIENT RUNWAY TO STOP		

ELECTRICAL FIRE		(4B)
1)	MASTER SWITCH	OFF
2)	GARMIN 7XX	ON AND SET TO SYNTHETIC VISION
NOTE USE FIRE EXTINGUISHER IF REQUIRED		
3)	CB'S	CHECKED/PULLED
4)	ELECTRICAL SWITCHES	OFF
5)	MASTER SWITCH	ON
6)	CB & SWITCHES FOR <u>ESSENTIAL ELECTRICS ONLY</u>	ON ONE AT A TIME UNTIL FAULTY CIRCUIT IS LOCATED
7)	CB & SWITCH FOR FAILED UNIT	OFF
8)	LAND AS SOON AS PRACTICAL	

RPM UNDERSPEED		(4C)
1)	POWER	REDUCED
2)	MIXTURE	RICH
3)	PROP CONTROL	ADJUST AS REQUIRED
IF PROP MOVES TO COARSE		
4)	MIXTURE	ICO
5)	PROP CONTROL	FEATHER
6)	ENGINE SECURING PROCEDURE	COMPLETE (1B)
NOTE PROP WILL MOVE TO COARSE (DECREASED RPM) IF ENGINE OIL PRESSURE IS LOST		

RPM OVERSPEED		(4D)
1)	POWER	REDUCED
2)	PITCH	NOSE UP TO REDUCE SPEED QUICKLY
3)	AIRSPEED	REDUCED TO APPROXIMATELY 130 KTS
4)	PROP (IF PROP SPEED CANNOT BE KEPT UNDER 2575RPM)	FEATHERED
5)	IF PROP FEATHERS	COMPLETE ENGINE SECURING PROCEDURE
NOTE IF PROP WILL NOT FEATHER, DO NOT SHUT DOWN ENGINE.		
NOTE WITH IAS BELOW 130 KTS MINIMAL POWER SHOULD BE AVAILABLE WITHOUT EXCEEDING 2575 RPM		



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CROSSFEED		(5A)
1)	FUEL SELECTOR (INOP ENGINE)	EITHER TANK
2)	FUEL BOOST CB (INOP ENGINE)	IN
3)	EMERGENCY FUEL PUMP (INOP ENGINE)	ON
4)	CROSSFEED	ON
5)	FUEL SELECTOR (OPERATIVE ENGINE)	OFF
6)	FUEL BOOST CB (OPERATIVE ENGINE)	PULLED
7)	EMERGENCY FUEL PUMP (OPERATIVE ENGINE)	OFF

ALTERNATOR FAILURE		(5B)
NOTE IT IS POSSIBLE TO HAVE AN ALTERNATOR INOP LIGHT AND STILL HAVE THE ALTERNATOR FUNCTIONING.		
1)	SYSTEM VOLTAGE (IF EQUIPPED)*	NOTED
2)	MASTER SWITCH (OPPOSITE TO INOP LIGHT)	OFF
*IF EQUIPPED AND VOLTAGE DROPS FROM 28 VOLTS TO 24 VOLTS (APPROXIMATELY) THE ALTERNATOR HAS FAILED. PROCEED TO STEP 3-6. IF THE VOLTAGE REMAINS AT APPROXIMATELY 28 VOLTS THE INOP LIGHT IS FAULTY AND ALTERNATOR OUTPUT WILL BE NORMAL. PLACE BOTH SIDES OF MASTER SWITCH BACK ON AND CONTINUE NORMAL OPERATIONS, OCCASIONALLY CHECKING ALTERNATOR VIA THE ABOVE PROCEDURE TO ENSURE THAT IT IS CONTINUING TO OPERATE.		
3)	ELECTRICS	REDUCE
4)	MASTER SWITCH (APPROPRIATE SIDE)	OFF
5)	CB'S	CHECK AND RESET
6)	MASTER SWITCH (APPROPRIATE SIDE)	BACK ON
NOTE IF ALTERNATOR IS FUNCTIONING NORMALLY REINSTATE ELECTRICAL LOAD. **NOTE** IF ALTERNATOR IS NOT OPERATING TURN THE APPROPRIATE SIDE OF THE MASTER OFF AND CONTINUE WITH REDUCED ELECTRICAL LOAD---LAND AS SOON AS PRACTICAL.		
DUAL ALTERNATOR FAILURE		
1)	ELECTRICS	REDUCE
2)	COMPLETE STEPS 4 TO 6 ONE ALTERNATOR AT A TIME	
3)	LAND AS SOON AS PRACTICAL	

EMERGENCY LANDING GEAR EXTENSION		(5C)
1)	AIRSPED MAX	128 KTS*
2)	GEAR SELECTOR	DOWN
3)	EMERG. GEAR COVER	OPEN
4)	EMERG. GEAR EXTENDER	EXTEND
5)	EXTENDER HANDLE PUMP UNTIL 3 GREEN & THE HANDLE RETURNS TO NEUTRAL. (APPROXIMATELY 50 STROKES)	PUMPED – FOR BEST RESULTS PUMP RAPIDLY
*ON POST 1978 MODELS THE LANDING GEAR VLE IS 153KIAS **NOTE** SIDE SLIPS CAN HELP LOCK THE MAINS. BECAUSE NOSE EXTENDS AGAINST AIRFLOW SLOW TO 90KTS, REDUCE POWER, FULL FLAPS AND NOSE DOWN ATTITUDE, THEN PUMP RAPIDLY. REMEMBER HYDRAULIC RESERVOIR HAS A LIMITED QUANTITY. IF UNSUCCESSFUL COMPLETE LANDING WITH TWO GEARS.		



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GEAR UP LANDING		(6A)
1)	FUEL	BURN OFF IF POSSIBLE
2)	PASSENGERS	BRIEFED
3)	NORMAL LANDING CHECKLIST	COMPLETE
4)	GEAR SELECTOR	UP
5)	AUTO PILOT	OFF
6)	MASTER SWITCH (DAYTIME)	OFF
WHEN LANDING IS ASSURED		
7)	MIXTURES	ICO
8)	PROP CONTROLS	FEATHER
9)	FIREWALL FUEL SHUTOFF	OFF
10)	FUEL SELECTORS	OFF
11)	MAGS	OFF
AT FULL STOP		
12)	EVACUATION DRILL	COMPLETE (1A)

AIR START (UNFEATHERING PROCEDURE)		(6B)
1)	FUEL SELECTOR	INBOARDS OR AS REQUIRED
2)	FUEL BOOST CB	IN
3)	MAGS	ON
4)	THROTTLE	OPEN ½"
5)	PROP	FORWARD TO MATCH OTHER CONTROL
6)	MIXTURE	SLOWLY FORWARD
7)	STARTER ENGAGED UNTIL PROP UNFEATHERS	
8)	PROP PULL BACK TO LOW RPM THROUGH 1000 RPM TARGET 2000 RPM	
9)	THROTLES REDUCED TILL WARM	2000 RPM MAX
10)	ALTERNATOR	ON
11)	ENGINE INSTRUMENTS	CHECK
12)	AFTER START CHECKS	COMPLETE

ENGINE ROUGHNESS		(6C)
1)	EMERGENCY FUEL PUMPS	ON
2)	MIXTURES	ADJUST
3)	ALTERNATE AIR	AS REQUIRED
4)	FUEL	SWITCH TANKS
5)	MAGS	CHECK
6)	REDUCE POWER	AS REQUIRED
***NOTE** IF THE ENGINE OPERATES SMOOTH ON ONE MAG LEAVE THE DEFECTIVE MAG OFF AND LAND AS SOON AS PRACTICAL.		



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ASYMMETERIC FLAPS (DUKE)		(7A)
1)	FLAP SWITCH	OFF
2)	AILERON	AS REQUIRED TO MAINTAIN WINGS LEVEL
3)	FLAP SWITCH	UP IN INCREMENTS TO ACHIEVE SYMMETRY
4)	FLAP SWITCH	OFF
5)	FLAP CONTROL CB	PULLED
NOTE DO NOT RESET THE FLAP CONTROL CB UNTIL THE AIRCRAFT HAS LANDED AND THE PROBLEM HAS BEEN RESOLVED, AS RESETTING THE SYSTEM AND ACTIVATION OF THE FLAP SELECTOR SWITCH WILL ENABLE ANOTHER OPERATION ALLOWING THE ASYMMETRIC FLAP DIFFERENTIAL TO BECOME MORE SEVERE.		

*See Additional Note Below**

FLAP SYSTEM MALFUNCTION (CALCO)		(7B)
ANNUCIATOR LIGHT ON		
1)	FLAP SELECTOR	REPOSITION SLIGHTLY
2)	IF FLAPS MOVE	REPLACE AMPLIFIER PRIOR TO NEXT FLIGHT
3)	IF FLAPS DO NOT MOVE	CHECK FOR SPLIT FLAP
4)	IF FLAPS ARE SPLIT	PULL FLAP MOTOR CB AND LAND IN THIS CONDITION
5)	IF FLAPS ARE NOT SPLIT	PULL AND RESET FLAP MOTOR CB
6)	IF FLAPS STILL DO NOT OPERATE	PULL FLAP MOTOR CB AND LAND IN THIS CONDITION
FLAPS FAIL TO RESPOND TO FLAP SELECTOR (WITH FLAP INDICATOR POINTING TO ZERO)		
1)	FLAP CONTROL CB	PULL AND RESET
2)	IF THE INDICATOR REMAINS OFF	THE FLAP CONTROL IS INOPERATIVE AND FLAPS CANNOT BE REPOSITIONED
3)	IF INDICATOR SHOWS FLAP POSITION	USE FOLLOWING CHECKLIST
FLAPS FAIL TO RESPOND TO FLAP SELECTOR (WITH FLAP INDICATOR POINTING TO FLAP POSITION)		
1)	FLAP TEST SWITCH	PUSH
2)	IF ANNUNCIATOR FAILS TO LIGHT	FOLLOW "ANNUNCIATOR LIGHT ON" CHECKLIST
3)	IF ANNUNCIATOR LIGHTS	PULL AND RESET FLAP MOTOR CB
4)	IF FLAPS FAIL TO RESPOND	A FLAP DRIVE FAULT MAY EXIST. FURTHER EFFORT TO REPOSITION FLAPS MAY CAUSE DAMAGE
NOTE IF FLAPS ARE NOT FULLY RETRACTED LAND AS SOON AS PRACTICAL, BEING CAREFUL NOT TO EXCEED THE FLAP SPEED. IF FLAP SYMMETRY CANNOT BE ACHIEVED, LAND AS SOON AS POSSIBLE, PLANNING THE CIRCUIT TO MAKE ALL TURNS TOWARD THE SIDE WITH THE FLAP THAT IS UP. IF A CROSS WIND IS PRESENT, LAND WITH THE WING WITH THE FLAP THAT IS UP, ON THE UP WIND SIDE.		



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SINGLE ENGINE LANDING		(8A)
1)	ENGINE SECURING PROCEDURE	COMPLETE (1B)
2)	HYDRAULIC PUMP	CHECK
3)	SEAT BELTS	SECURE
4)	HEATER	FAN
OPERATIVE ENGINE		
5)	EMERGENCY FUEL PUMP	ON
6)	MIXTURE	RICH
7)	PROP	FORWARD
8)	FUEL SELECTOR	INBOARD
9)	CROSSFEED	OFF
10)	COWL FLAPS	AS REQUIRED
11)	AIRSPEED MAINTAIN	116 KTS OR ABOVE
GLIDESLOPE – 1 DOT		
12)	FLAPS	15°
GLIDESLOPE – ON SLOPE		
13)	GEAR	DOWN
LANDING ASSURED		
14)	FLAPS	FULL
NOTE WHEN FLYING AN ILS OR 3 DEGREE GLIDE SLOPE, FLY THE AIRCRAFT SLIGHTLY ABOVE SLOPE AND SLIGHTLY ABOVE NORMAL AIRSPEED WITH THE LANDING GEAR DOWN AND 15 DEGREES FLAP. IF RUNWAY LENGTH IS A CONCERN, MORE FLAP MAY BE USED ON SHORT FINAL WHEN LANDING IS ASSURED. MAINTAIN POSITIVE ENERGY! REMEMBER YOUR THREE SOURCES OF ENERGY.		

PRIMARY FLIGHT DISPLAY		(8B)
1)	BLANK SCREEN	FLY STBY HORIZON
2)	MFD (IF INSTALLED)	SELECT FLIGHT DISPLAY
3)	GARMIN 7XX	SELECT SYNTHETIC VISION
4)	GARMIN 530/430	SELECT NAV DISPLAY AS REQUIRED
5)	TROUBLESHOOT	CHECKLISTS

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**SINGLE PILOT OPERATIONS REQUIRE THE PILOT TO COMPLETE BOTH
PM/FO AND PF/CAPTAIN CHECKS**

EMERGENCY SPEEDS

V _{mc}	72 KTS
V _{xse}	104 KTS
V _{yse}	107 KTS
V _a	159 KTS
V _{ne}	236 KTS

OPERATIONAL SPEEDS

V _{barrier}	95 KTS
V _x	84 KTS
V _y	101 KTS
V _{le} *	128 KTS
V _{lo} *	128 KTS
V _{fe} (0-25°)**	153 KTS
V _{fe} (25-40°)	130 KTS

***On post 1978 models the landing gear V_{le} and V_{lo} is 153 KTS**

****On post 1978 models V_{fe} 0-25° is 162 KTS**